



CSA1507 - CLOUD COMPUTING AND BIG DATA ANALYTICS



Cloud Security Fundamentals: Protecting Data and Applications

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Introduction



Cloud computing allows users to store data and run applications over the internet. It provides easy access to computing resources from anywhere.

- Cloud security protects cloud systems, applications, and stored data from threats. It helps prevent unauthorized access, cyberattacks, and data loss.
- The main purpose of cloud security is to ensure Confidentiality, Integrity, and Availability (CIA). These three principles keep information safe, accurate, and accessible.
- Cloud service providers use technologies such as encryption, firewalls, and Identity and Access Management (IAM).
- These security measures protect data and control user access.



Objectives



- Understand cloud security concepts.
- Learn cloud security architecture.
- Identify common threats.
- Explain the Shared Responsibility Model.
- Explore IAM, encryption, and data protection.
- Discuss industry applications and future trends.



Why it's important



Protects Sensitive Data

- Cloud security safeguards personal, financial, and business information from unauthorized access.

Prevents Cyber Attacks

- It helps defend against malware, ransomware, phishing, and other online threats.

Ensures Business Continuity

- Regular backups and disaster recovery help keep services running during failures.

Maintains Customer Trust

- Strong security measures protect customer information and increase confidence in cloud services.

Supports Legal Compliance

- It helps organizations meet data protection laws and industry security standards.



Common Security Threats



Data Breaches

- Sensitive information may be exposed when attackers gain unauthorized access to cloud data.

Malware and Ransomware

- Malicious software can infect cloud systems, steal data, or lock files until a ransom is paid.

Phishing Attacks

- Fake emails or websites trick users into revealing passwords and other confidential information.

Weak Passwords

- Simple or reused passwords make it easier for attackers to access cloud accounts.

Insider Threats

- Employees or trusted users may intentionally or accidentally expose sensitive data.



Types of Cloud Security



1. Network Security

Protects cloud networks from unauthorized access and cyber attacks.

Uses firewalls, VPNs, and intrusion detection systems.

2. Data Security

Protects data stored and transmitted in the cloud.

Uses encryption, backups, and data loss prevention (DLP).

3. Identity and Access Management (IAM)

Controls who can access cloud resources.

Uses authentication, authorization, and Multi-Factor Authentication (MFA).

4. Application Security

Protects cloud-based applications from vulnerabilities and attacks.

Includes regular updates, secure coding, and vulnerability testing.



Data Protection and Encryption



Data Protection

- Data protection ensures that sensitive information is safe from unauthorized access, loss, or theft.
- It uses backups, access controls, and security policies to maintain data confidentiality and integrity.

Encryption

- Encryption converts readable data into an unreadable format using a secret key.
- Only authorized users with the correct decryption key can access the original information.



Real world application



Healthcare

Protects patient records and medical data from unauthorized access.
Ensures secure communication between hospitals and healthcare providers.

Banking and Finance

Secures online banking transactions and customer account information.
Prevents fraud, data theft, and cyberattacks.

E-Commerce

Protects customer personal information and payment details.
Provides secure online shopping and digital payment services.

Education

Secures student records and online learning platforms.
Enables safe access for students, teachers, and administrators.



Challenges



Unauthorized users may gain access to sensitive cloud data, leading to data breaches. This can result in financial loss and damage to an organization's reputation.

Incorrect cloud configurations can expose systems and applications to cyber threats.

Regular security reviews and proper settings help reduce this risk.

Insider threats occur when employees or trusted users misuse cloud resources.

Strong access controls and continuous monitoring help prevent such incidents.

Organizations must comply with security regulations and data protection laws.

Failure to meet compliance requirements can result in legal and financial penalties.

Data loss may occur due to cyberattacks, hardware failures, or accidental deletion.

Regular backups and disaster recovery plans help restore lost information.

Future Scope

- Artificial Intelligence (AI) will improve cloud security by detecting and preventing cyber threats automatically. AI-based systems can respond to attacks faster and more accurately.
- Zero Trust Security will become a standard approach for protecting cloud environments. Every user and device will be verified before accessing cloud resources.
- Multi-Factor Authentication (MFA) and biometric authentication will become more advanced. These methods will provide stronger protection against unauthorized access.
- Cloud providers will use automation to monitor systems and respond to security incidents. Automated tools will reduce human errors and improve security management.

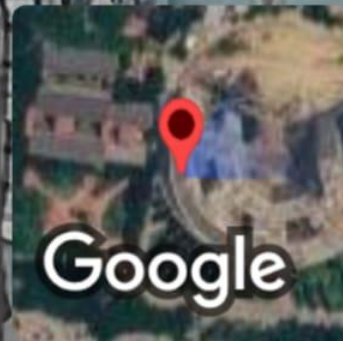
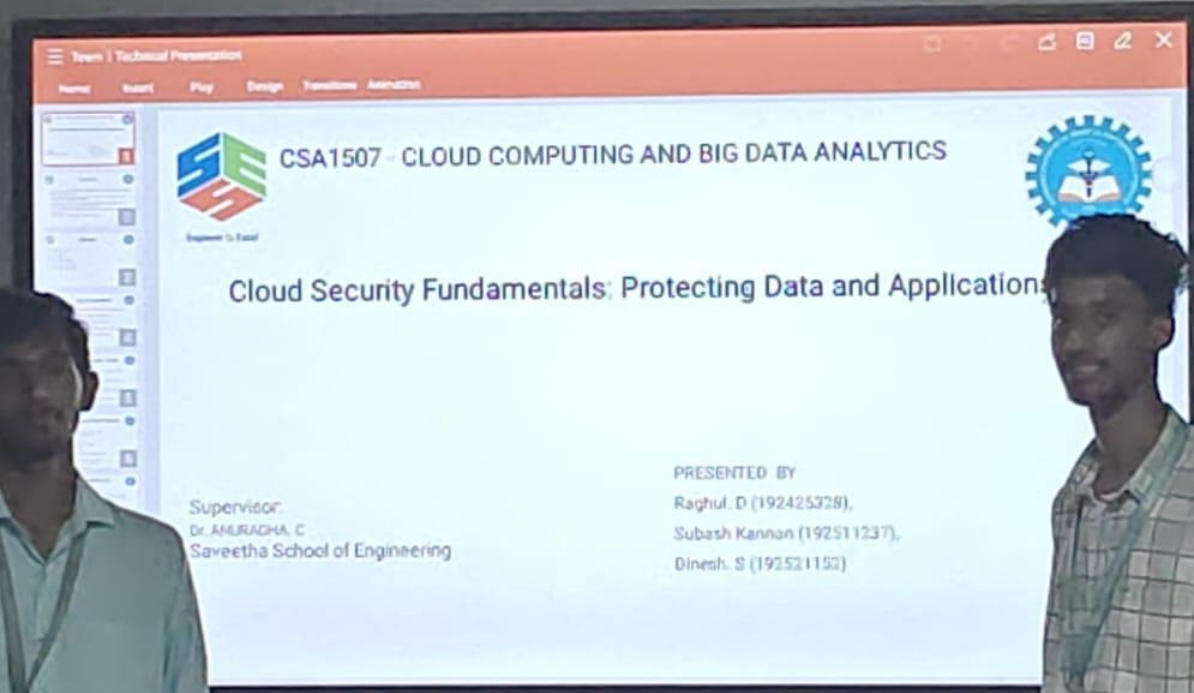


Conclusion



- Cloud security protects data, applications, and cloud services from cyber threats. It helps ensure safe and reliable cloud computing.
- Technologies like encryption, IAM, and MFA improve cloud security. They prevent unauthorized access and protect sensitive information.
- Following security best practices reduces risks and improves data protection. This helps organizations use cloud services with confidence.
- As technology advances, cloud security will continue to become stronger and smarter. It will play a key role in the future of cloud computing.

THANK YOU



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